# Chapter 3: Chemical Kinetics

Formula Sheet + Tips & Tricks

## Formula Sheet

### I. Rate of Reaction

**Average Rate**

$$\text{Average rate} = \frac{\Delta[\text{conc.}]}{\Delta t} = \frac{C_2 - C_1}{t_2 - t_1}$$

Change in concentration of reactant/product over a time interval.

**Instantaneous Rate**

$$\text{Instantaneous rate} = -\frac{d[R]}{dt}$$

Rate of change of concentration of a reactant at a particular instant.

**General Rate Expression**

$$\text{Rate} = -\frac{1}{a}\frac{d[A]}{dt} = -\frac{1}{b}\frac{d[B]}{dt} = \frac{1}{c}\frac{d[C]}{dt} = \frac{1}{d}\frac{d[D]}{dt}$$

For a general reaction $aA + bB \rightarrow cC + dD$; reactant terms carry a minus sign, product terms a plus sign.

### II. Rate Law and Order

**Rate Law**

$$\text{Rate} = k[A]^x[B]^y$$

$k$ = rate constant (specific to the reaction, depends on temperature); $x, y$ = orders w.r.t. A and B, determined experimentally (not from stoichiometry).

**Order of Reaction**

$$\text{Order} = x + y$$

Sum of the powers of concentration terms in the experimentally determined rate law.

### III. Integrated Rate Equations & Half-Life

### Zero Order – Rate Constant

$$k = \frac{A_0 - A_t}{t}$$

$A_0$ = initial concentration; $A_t$ = concentration at time $t$.

### Zero Order – Half Life

$$t_{1/2} = \frac{A_0}{2k}$$

Half-life is directly proportional to initial concentration for a zero order reaction.

### First Order – Rate Constant

$$k = \frac{2.303}{t} \log_{10} \frac{A_0}{A_t}$$

Also written in exponential form: $[A]_t = [A]_0\, e^{-kt}$.

### First Order – Half Life

$$t_{1/2} = \frac{0.693}{k}$$

Independent of initial concentration – a hallmark of first order kinetics.

## IV. Activation Energy & Arrhenius Equation

### Activation Energy

$$E_a = E_{\text{threshold}} - E_{(\text{average reactants})}$$

Additional energy required to raise reactant molecules to the threshold energy so the reaction can occur.

### Enthalpy of Reaction

$$\Delta H = E_a - E_b$$

$E_a$ = activation energy of forward reaction; $E_b$ = activation energy of backward reaction.

### Arrhenius Equation

$$k = A\, e^{-E_a/RT}$$

$A$ = Arrhenius (frequency) factor; $E_a$ = activation energy; $R$ = gas constant; $T$ = temperature (K).

### Arrhenius Equation (log form)

$$\log k = \log A - \frac{E_a}{2.303RT}$$

Straight-line form: plotting $\log k$ vs. $1/T$ gives slope $= -\frac{E_a}{2.303R}$ and intercept $= \log A$.

### Arrhenius Equation (two-temperature comparison)

$$\log \frac{k_2}{k_1} = \frac{E_a}{2.303R}\left(\frac{1}{T_1} - \frac{1}{T_2}\right)$$

Used to find $E_a$ given rate constants at two different temperatures.

## Tips, Tricks & Hacks

- **Master units table**: Rate of reaction – mol $L^{-1}s^{-1}$; $k$(zero order) – mol $L^{-1}s^{-1}$; $k$(first order) – $s^{-1}$; $k$(second order) – L $mol^{-1}s^{-1}$; $k$($n$th order) – $mol^{1-n}L^{n-1}s^{-1}$; half-life – s (or min); activation energy – J $mol^{-1}$.
- If a question gives "% completed" instead of concentrations: e.g., 80% completed means $A_t = 0.2\,A_0$ (20% remains). For first order: 50% completed $\Rightarrow A_0/A_t = 2$; 75% $\Rightarrow$ 4; 90% $\Rightarrow$ 10; 99% $\Rightarrow$ 100.
- "Time for concentration to fall to 1/4 of initial" $= 2 \times t_{1/2}$ (two successive half-lives), directly – no need to compute $k$ first.
- **Order vs. molecularity**: order is an experimental quantity and can be zero, fractional, or negative; molecularity is always a whole number (never zero or fractional) and equals the number of colliding species in an elementary step. For a simple (single-step) reaction, order = molecularity.
- Pseudo first order reactions: a higher-order reaction that behaves like first order because one reactant is in large excess (e.g., acid hydrolysis of esters, inversion of cane sugar).
- Collision theory conditions for effective reaction: (1) sufficient/optimum energy, (2) proper orientation of colliding molecules.
- A catalyst provides an alternate pathway with *lower* activation energy – it does not change $\Delta H$ of the reaction.